

A
Project Report On
**A Study on Cybersecurity Awareness among Undergraduate and Postgraduate
College Students in the Varale Area**



Savitribai Phule Pune University, Pune

**In Partial Fulfillment of
Master of Computer Application
(MCA-II, Sem-III)**

Submitted by

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Under the Guidance of

Prof. Kiran Shejul

Through



**Dr. D. Y. Patil Educational Federation's
DR. D. Y. Patil Institute of Management and Entrepreneur
Development
2025-26**



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This is to certify that **Mr. Akshay Sanjay Gunjal** has successfully completed his research project work entitled “**A Study on Cybersecurity Awareness among Undergraduate and Postgraduate College Students in the Varale Area**” in partial fulfilment of MCA – II SEM –III (RP641RP) Research Project for the year 2025-2026. He/she has worked under our guidance and direction.

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DECLARATION

I, **Akshay Sanjay Gunjal**, a student of Master of Computer Applications (MCA) at Savitribai Phule Pune University (SPPU), hereby declare that the research paper titled “**A Study on Cybersecurity Awareness among Undergraduate and Postgraduate College Students in the Varale Area**” is my original and independent work carried out under the guidance of **Prof. Kiran Shejul Sir**.

This research paper has not been submitted to any other university or institution for the award of any degree, diploma, or certificate. The information, analysis, and interpretations presented in this report are the result of my own study, supported by duly acknowledged sources and references.

I take full responsibility for the authenticity and integrity of the contents presented in this research paper.

Signature:

Name: **Mr. Akshay Sanjay Gunjal**

ACKNOWLEDGEMENT

I express my sincere gratitude to **Prof. Kiran Shejul** Sir for his invaluable guidance, encouragement, and constant support throughout the completion of this research paper entitled “**A Study on Cybersecurity Awareness among Undergraduate and Postgraduate College Students in the Varale Area** ” His expert insights, constructive feedback, and unwavering motivation have been instrumental in shaping the quality and direction of this study.

I would also like to extend my heartfelt thanks to Savitribai Phule Pune University (SPPU) for providing me with the opportunity and academic platform to undertake this research as part of my curriculum requirements.

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Mr. Akshay Sanjay Gunjal

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LIST OF ACRONYMS

Acronyms	Full Form
AI	Artificial Intelligence
IoT	Internet of Things
UG	Undergraduate
PG	Postgraduate
ICT	Information and Communication Technology
MFA	Multi-Factor Authentication
VPN	Virtual Private Network
ANOVA	Analysis of Variance
SD	Standard Deviation
N	Total Sample Size
H0	Null Hypothesis
H1	Alternative Hypothesis
PCSPS	Perceived Cybersecurity Practice Scale
HAIS-Q	Human Aspects of Information Security Questionnaire

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1.1 INTRODUCTION

The digital age has fundamentally reshaped academic and personal environments, establishing cybersecurity as an indispensable component of digital literacy and everyday life. This transformation involves heavy reliance on connected devices and online platforms for crucial activities, including online learning, communication, and financial transactions. College students, in particular, represent a demographic of highly active technology users, consequently placing them at high risk for an increasing array of cyber threats.

These risks are multifaceted, ranging from common tactics like phishing and the infiltration of harmful software (malware) to more complex issues such as identity theft and malicious social engineering attacks designed to manipulate individuals into compromising data. Even in the face of a significant and continuous rise in cybercrime and verified instances of data breaches, many students still demonstrate an alarming lack of sufficient awareness. This deficiency often manifests as a pronounced knowledge-behavior gap, where theoretical understanding fails to translate into consistent, secure online practices, making them easy targets for exploitation.

Therefore, this local research aims to precisely evaluate the current level of cybersecurity awareness among the undergraduate and postgraduate college students in the Varale area, providing localized, empirical data necessary to pinpoint areas where training is failing and recommend evidence-based improvements. This study directly addresses the necessity of ensuring that digital usage is accompanied by adequate protection, which is vital for the safety of the individual and the integrity of academic systems.

1.2 STATEMENT OF THE PROBLEM

The increasing and ubiquitous reliance on technology for all aspects of modern academic life including blended learning environments, digital communication, and financial transactions in escapably positions college students as frequent and vulnerable targets of cybercrimes. Students, while digitally native, often exhibit poor cyber hygiene, lacking the necessary practical knowledge regarding safe online practices, robust password management, and comprehensive data protection protocols. This negligence often includes behaviors such as reusing passwords, ignoring software updates, and clicking on suspicious links.

This problem is compounded by a well-documented global phenomenon: the knowledge-behavior gap. Previous research consistently confirms a pervasive crisis where students may possess superficial awareness of cyber threats but fail to translate this knowledge into secure behavior. Students often consciously choose to ignore safety measures due driven by convenience or a flawed risk assessment, perceiving the immediate effort of security measures as higher than the abstract threat of a cyberattack.

The challenge addressed by this research, therefore, is the specific lack of localized data to quantify the true extent of this knowledge-behavior gap within the Varale area college student population. Without empirical evidence specific to their environment, local institutions cannot develop effective, targeted interventions. This study is essential to measure the disparity between local students' high-frequency digital tool use and their awareness of the essential, practical cybersecurity measures needed to protect themselves from imminent threats like phishing and data theft.

1.3 OBJECTIVES OF THE RESEARCH

The research project is structured around four distinct and measurable objectives designed to systematically investigate the cybersecurity landscape within the local student population and provide actionable recommendations:

- To evaluate the current level of cybersecurity awareness among college students.

This objective aims to quantify the baseline knowledge held by undergraduate and postgraduate students, specifically assessing their theoretical understanding of core cybersecurity concepts, terminology, and standard protective measures. This establishes the intellectual starting point of students regarding online safety.

- To identify common cyber threats that students face in their academic and personal digital activities.

This involves analyzing self-reported data to determine the most prevalent risks and vulnerabilities encountered by students (e.g., phishing attempts, malware infections, social engineering, or identity theft). By localizing these common threats, the research ensures that subsequent policy recommendations are highly relevant and targeted to the students' actual exposure environment.

- To examine the connection between students' digital habits and their cybersecurity practices.

This objective directly investigates the existence and magnitude of the knowledge-behavior gap. It involves analyzing the correlation between a student's high-frequency digital reliance (their habits, like daily internet use) and their engagement in secure behaviors (their practices, such as strong password management and source verification). The goal is to determine where awareness is failing to translate into consistent, practical safety actions.

- To recommend strategies for improving cybersecurity awareness through effective educational programs and training.

1.4 HYPOTHESIS OF THE STUDY

The research is guided by a set of formal hypotheses which serve as educated predictions to be tested using the empirical data gathered from the student population. These hypotheses address the core question of whether cybersecurity awareness is sufficient and uniform across different student groups:

- Null Hypothesis (H0): This hypothesis represents the starting assumption that there is no meaningful impact or variation in the observed data. The Idea: There is no significant level of cybersecurity awareness among college students that translates to effective protection, or conversely, there is no significant difference in cybersecurity awareness based on demographic or academic factors (such as education level, gender, or course). This would suggest that current training methods are universally ineffective or that awareness levels are uniform but low.
- Alternative Hypothesis (H1): This hypothesis is the prediction the research aims to prove, suggesting that observed differences or levels are not due to chance. The Idea: There is a significant level of cybersecurity awareness among college students, and/or there is a significant difference in cybersecurity awareness based on demographic or academic factors. For example, H1 would be supported if postgraduate students exhibited demonstrably higher knowledge scores than undergraduates, or if the overall level of awareness was insufficient, manifesting as the quantified knowledge-behavior gap.

1.5 SIGNIFICANCE OF THE STUDY

The significance of this research project extends across several critical levels, providing practical value to key stakeholders:

- **For Local College Management and Policymakers (The Institutional Value)**
This study is crucial because it provides localized, empirical data evidence specific to the students and environment of the Varale area. Global or national studies often miss local nuances, but this research offers precise, factual insights into student behavior and vulnerability. This information is indispensable for college management and policymakers to formally recognize and quantify the urgent need for organized awareness programs. The findings will help institutions move beyond generic safety guidelines to create effective, targeted cybersecurity policies and intervention strategies that address quantified local risks, such as password reuse and phishing exposure.
- **For Educators and Faculty (The Pedagogical Value)**
The study offers crucial pedagogical insights to educators. By examining the knowledge-behavior gap, the research highlights where current training methods are failing to translate awareness into secure action. The findings guide faculty in designing effective, targeted cybersecurity training modules that prioritize experiential and hands-on learning methods students prefer. This shift ensures that instruction focuses on promoting practical cyber hygiene and developing the "perceived behavioral control" necessary for consistent secure behavior.
- **For Future Students and the Job Market (The Societal Value)**
Ultimately, this research serves to protect the future digital workforce. College students, as the most active technology users, are continually preparing for an increasingly interconnected job market where cybersecurity competency is essential. By improving digital literacy and safety practices now, the study helps reduce students' vulnerability to personal and academic cybercrime. Moreover, the findings could aid in developing national strategies to improve digital literacy and safety among young people, ensuring they possess the foundational security skills required for success in the global economy.

2 REVIEW OF LITERATURE

The literature confirms that growing dependence on technology has increased student vulnerability to online threats. While awareness is high, secure behavior remains insufficient, establishing the persistent **knowledge-behavior gap**.

Theme	Key Findings & Supporting Sources
The Knowledge-Behavior Gap	Students know basic concepts (password protection, antivirus use) but practice unsafe behaviors like reusing passwords and ignoring software updates. Postgraduates show higher theoretical knowledge but lack practical awareness. The problem is the failure to apply knowledge (flawed risk assessment), not a lack of information.
Localized Gaps and Vulnerability	Awareness is lower among students in rural or semi-urban areas like Varale due to lack of exposure and limited access to digital infrastructure. Over 60% of students in phishing simulations clicked on fraudulent links, indicating low awareness levels.
Training Effectiveness & Pedagogy	Interactive, activity-based training (workshops) results in better awareness retention compared to lecture-based formats. Literature calls for a shift to experiential, hands-on, and adaptive learning .
Psychological & Policy Factors	Perceived behavioral control (self-efficacy/confidence) is the most powerful predictor of safe behavior. Cybersecurity education should be integrated into the college curriculum , and there is a critical need to teach cyber laws and how to report incidents.

3 RESEARCH METHODOLOGY

3.1 RESEARCH APPROACH AND DESIGN

- Research Aim and Scope

The central goal of this research is highly pragmatic: to evaluate the level of cybersecurity awareness among the diverse population of college students (Undergraduate and Postgraduate) within the Varale area, systematically identify common risky behaviours exhibited by this demographic, and ultimately recommend practical, evidence-based interventions for improvement.

- Research Approach: Mixed-Methods Design

The study is founded upon a Mixed-Methods design. This approach is chosen specifically to capture the complexity of human-technology interaction and bridge the gap between "what is done" and "why it is done."

- Primary Focus (Quantitative): The design is primarily quantitative, relying on a structured questionnaire to generate objective numerical data. This allows for rigorous statistical testing of hypotheses and the calculation of proportions, frequencies, and scores related to awareness levels and unsafe practices.

- Secondary Focus (Qualitative): A necessary qualitative component (e.g., open comments or short interviews) is included to provide context to the statistical findings. This helps understand the underlying reasons, barriers, and behavioral motivations that explain the observed numerical results, particularly why students knowingly engage in risky actions.

- Design Type: Cross-sectional, Descriptive-Analytic Study

The research employs a Cross-sectional, descriptive-analytic study design.

Cross-sectional: Data is collected at one single point in time to measure the prevalence of awareness and specific behaviors across the student population.

- Descriptive-Analytic: The study describes the current state of awareness and analytically tests the associations between different variables (e.g., comparing scores between UG vs. PG students, or training exposure vs. practice scores).

- Population

- The target population comprises all undergraduate (UG) and postgraduate (PG) students enrolled in colleges located in the Varale area. This tight geographical and demographic focus ensures the generated data is directly actionable for local institutions and policymakers.

3.2 SAMPLING STRATEGY

- Sampling Strategy Rationale

To ensure the collected data accurately reflected the heterogeneity of the target academic population, a Stratified Random Sampling approach was initially recommended. This is vital to guarantee proportional representation across key variables, namely different education levels (UG/PG) and the specific colleges within the Varale area. Given resource constraints or lack of complete student lists, this method was practically implemented using a Quota Sampling technique. This ensured that the researcher targeted a specific, proportional number of participants from each stratum (e.g., MCA, Engineering, UG vs. PG)

- Sampling Criteria: The integrity of the local data relies on rigorous adherence to the following criteria:

- Inclusion Criteria:

Participants must be currently enrolled UG or PG students in one of the Varale-area colleges, be aged 17 or above 4, and actively willing to give informed consent. This strict focus ensures that the collected data directly relates to the study's specific local context and population.

- Exclusion Criteria (Implicit):

Non-students, alumni, and any individuals unable or unwilling to complete the questionnaire were excluded.

- Sample Size Achieved:

The data collection effort successfully concluded with a final analyzed sample size of $N = 159$ respondents.

3.3 DATA COLLECTION INSTRUMENT AND VARIABLES

- The Data Collection Instrument: Structured Questionnaire

The primary instrument for data collection was a structured, self-administered questionnaire. This tool was selected for its efficiency in gathering standardized quantitative data across a large and dispersed student population, ensuring consistency in measurement. The use of a fixed-response format allowed for easy numerical coding and subsequent statistical analysis. Crucially, the questionnaire was pilot tested with a small, representative group of students (from nearby, non-study colleges) prior to the main deployment. This step was essential to ensure the instrument's clarity, reliability, and face validity, minimizing ambiguity and verifying that the questions effectively measured the intended concepts.

- Operationalization of Variables

The measurement items within the questionnaire were designed to capture the relationship between explanatory and outcome variables:

- Independent Variables : These are the demographic and experiential factors assumed to influence cybersecurity outcomes.
 - Variables: Age, gender, education level (UG/PG), and prior cybersecurity training exposure.
 - Purpose: Used as categorical variables in inferential analysis (e.g., t-tests or ANOVA) to determine if significant differences exist in awareness or behavior between groups (e.g., comparing scores of UG vs. PG students).
- Dependent Variables: These are the measurable outcomes that reflect the levels of awareness and safety within the student body.
 - Variables: Cybersecurity knowledge score, cybersecurity behaviour/practice score, attitude towards cybersecurity, and intention to participate in training.
- Operationalization Examples:
 - Knowledge Score: Calculated as the sum of correct answers to factual multiple-choice questions (e.g., defining phishing).
 - Practice/Behavior Score: Measured via frequency-based scales (e.g., Likert-type scales such as "1 = Never" to "5 = Always") for specific safe behaviors, such as password creation methods and the habit of checking links. This provided the empirical data to quantify the knowledge-behavior gap.

3.4 DATA ANALYSIS PLAN

1. Preliminary Analysis and Reliability Check

- Data Integrity: Initial verification ensured that categorical data (e.g., College Name, Course) was standardized and that no incomplete or ambiguous entries were included.
- Reliability: For multi-item scales (such as those measuring attitudes or practices), calculating Cronbach's alpha (alpha) was planned to ensure the internal consistency of the scales, aiming for a generally accepted threshold of alpha 0.7.

2. Quantitative Analysis: Descriptive Statistics

- Frequency and Percentage Analysis: Calculating the frequency and percentage for categorical data (e.g., 60.4% daily internet use, 34.0% unsafe password behavior) to directly quantify awareness levels and risky practices .
- Mean Scores and Standard Deviations (SDs): Calculating the average score (Mean) and spread (SD) for continuous variables, such as cybersecurity knowledge and practice scores, to quickly determine the overall student sentiment and performance.

3. Inferential Analysis (Hypothesis Testing)

- Comparison of Mean Scores (t-tests/ANOVA): The planned use of independent samples t-tests or Analysis of Variance (ANOVA) was necessary to compare the mean scores of Dependent Variables (e.g., knowledge score) across different key Independent Variable groups (e.g., UG vs. PG students, Male vs. Female students). This determined if significant differences exist in awareness or behavior based on demographic factors (H1).
- Correlation Analysis (Planned): Analyzing the correlation (e.g., Pearson's r) between variables like "prior cybersecurity training" and "practice score" to examine the strength of the relationship and further address the knowledge-behavior gap

4 PROPOSED WORK AND EXECUTION

4.1 EXECUTION OF DATA COLLECTION

Target Group:

The primary respondents for this research were undergraduate (UG) and postgraduate (PG) students enrolled in colleges located within the Varale area. This group was chosen deliberately because they represent a highly active demographic in terms of digital engagement and technology usage. Students at both UG and PG levels frequently rely on the internet for academic activities, communication, and entertainment, making them ideal participants for a study focused on cybersecurity awareness. The inclusion of students from different academic levels also allowed for comparative analysis between varying degrees of exposure, knowledge, and experience with digital tools and security practices.

Distribution Rationale:

- **Accessibility:** Students could access the survey at their convenience using smartphones, laptops, or desktops.
- **Anonymity:** Responses were collected anonymously, encouraging participants to answer honestly without fear of judgment or bias.
- **Efficiency:** The online format facilitated faster data collection and minimized manual errors during data entry.
- **Instant Data Capture:** The use of Google Forms enabled automatic recording and organization of responses, which simplified subsequent data cleaning and analysis.

Final Data Set:

The data collection process was completed successfully within the planned timeframe. After eliminating incomplete and inconsistent responses, a total of 159 valid and complete responses were obtained. This dataset formed the foundation for the subsequent stages of analysis, providing meaningful insights into the cybersecurity awareness levels of college students in the Varale region. The response rate was considered satisfactory for the scope of this research, and the diversity among respondents ensured that the findings could be interpreted as broadly representative of the student community in the area.

4.2 DATA PREPARATION AND CODING

Once the data collection phase was completed, the next crucial step involved systematically preparing the gathered responses for descriptive analysis. Proper data preparation ensured the accuracy, consistency, and reliability of the findings derived from the dataset. Each stage of this process was carried out carefully to maintain data integrity and to ensure that the results accurately reflected the participants' responses.

Verification:

In the initial stage, all responses were thoroughly examined to verify their completeness and accuracy. Incomplete entries, duplicates, or inconsistent responses were identified and excluded to prevent any distortion in the results. This verification process also included checking whether participants had responded to all mandatory questions and whether their answers aligned logically with the question type. Additionally, categorical data such as demographic information including gender, academic level, and college affiliation was standardized to maintain uniformity across responses. For instance, minor variations in text entries (e.g., “B.Sc” and “BSc”) were normalized to a single format to facilitate consistent analysis. This step was essential to ensure that data processing and interpretation could be carried out smoothly without discrepancies.

Coding:

After verification and cleaning, the next step involved coding the data to convert qualitative and categorical responses into numerical or symbolic representations. Each response option in the multiple-choice questions was assigned a unique numerical code to make the dataset compatible with statistical tools and to simplify computation. This systematic coding allowed for direct calculation of frequency distributions and percentage counts, which were later used to perform descriptive analysis. Through this process, patterns such as the most common responses, the proportion of awareness levels, and comparative trends among different groups of students could be efficiently identified.

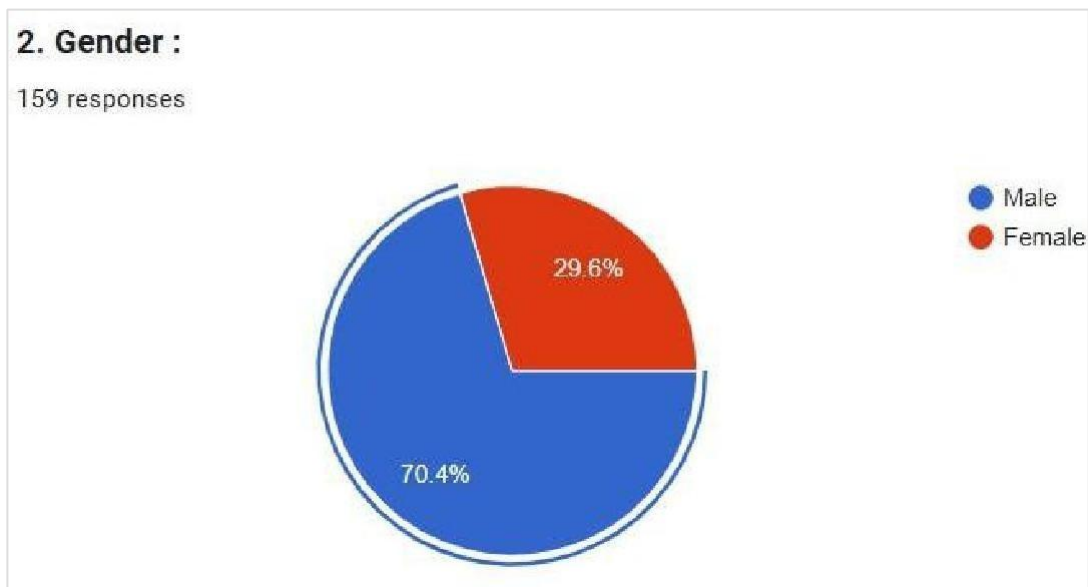
5 RESULT AND DISCUSSION

5.1 SAMPLE REPRESENTATION

1. Gender :

Gender	Count (N=159)	Percentage
Male	112	70.4%
Female	47	29.6%

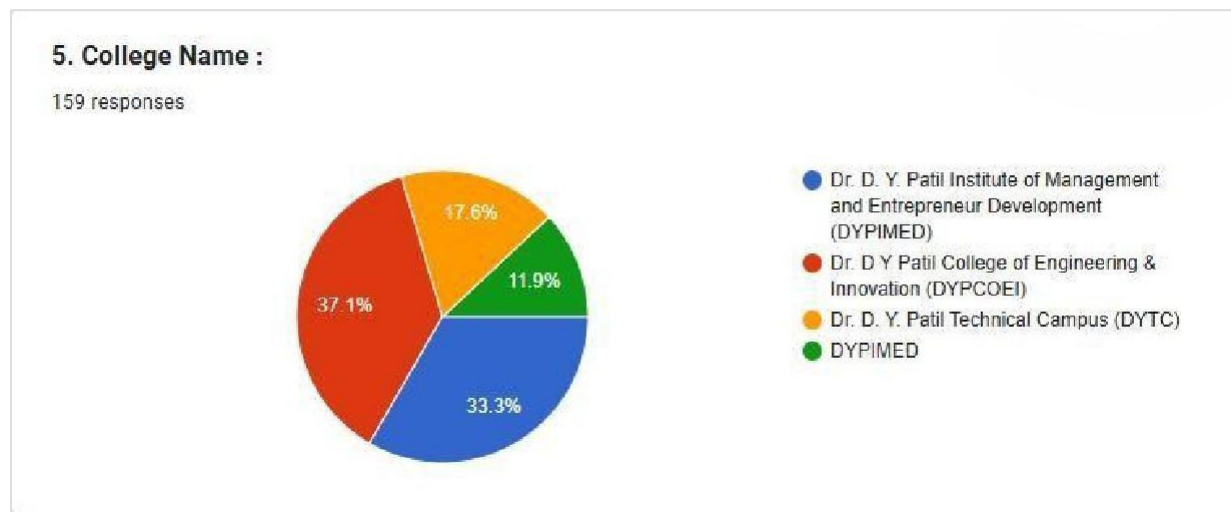
Pie Chart :



2. College :

College Name	Count (N=159)	Percentage
Dr. D. Y. Patil College of Engineering & Innovation (DYPCOEI)	59	37.1%
Dr. D. Y. Patil Institute of Management and Entrepreneur Development (DYPIMED)	53	33.3%
Dr. D. Y. Patil Technical Campus (DYTC)	28	17.6%
DYPIMED (Duplicate/Self-reported)	19	11.9%

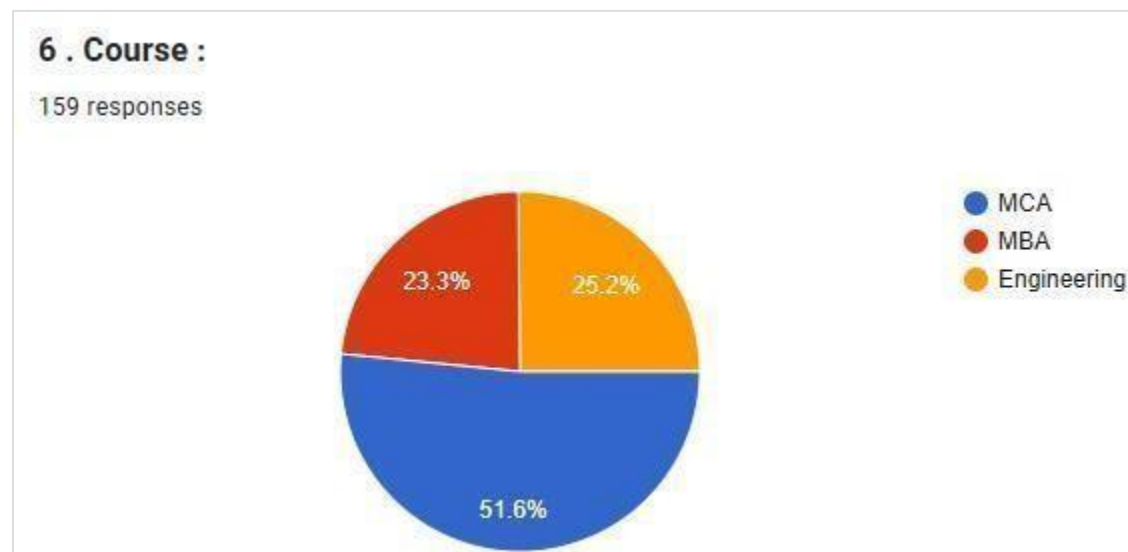
Pie Chart :



3. Course :

Course	Count (N=159)	Percentage
MCA	82	51.6%
Engineering	40	25.2%
MBA	37	23.3%

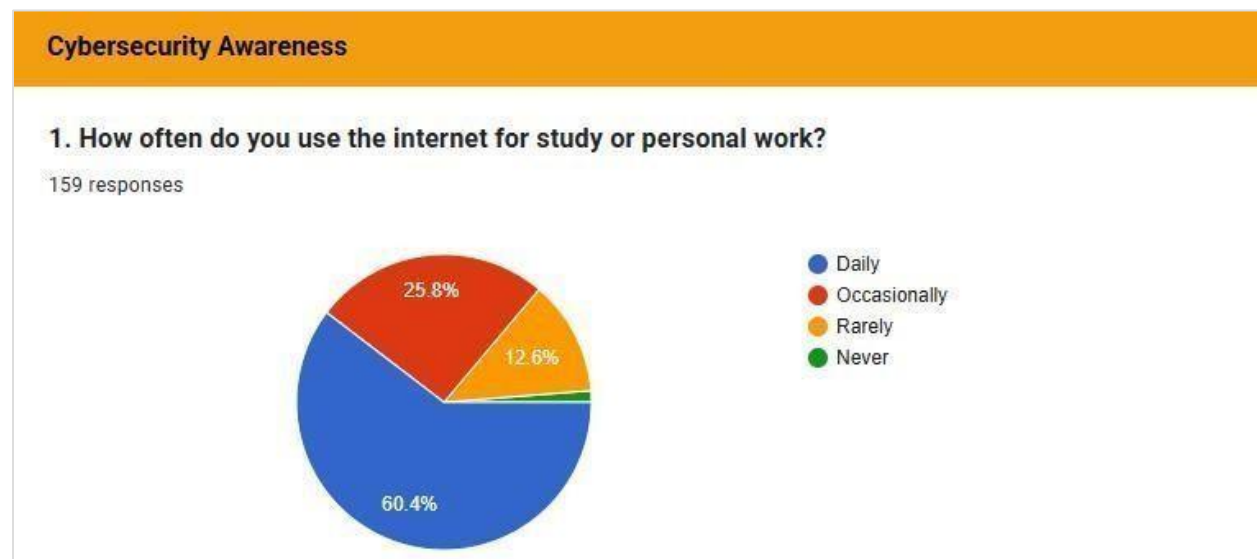
Pie Chart :



4. Usage :

Usage Frequency	Count (N=159)	Percentage
Daily	96	60.4%
Occasionally	41	25.8%
Rarely	20	12.6%
Never	2	1.3%

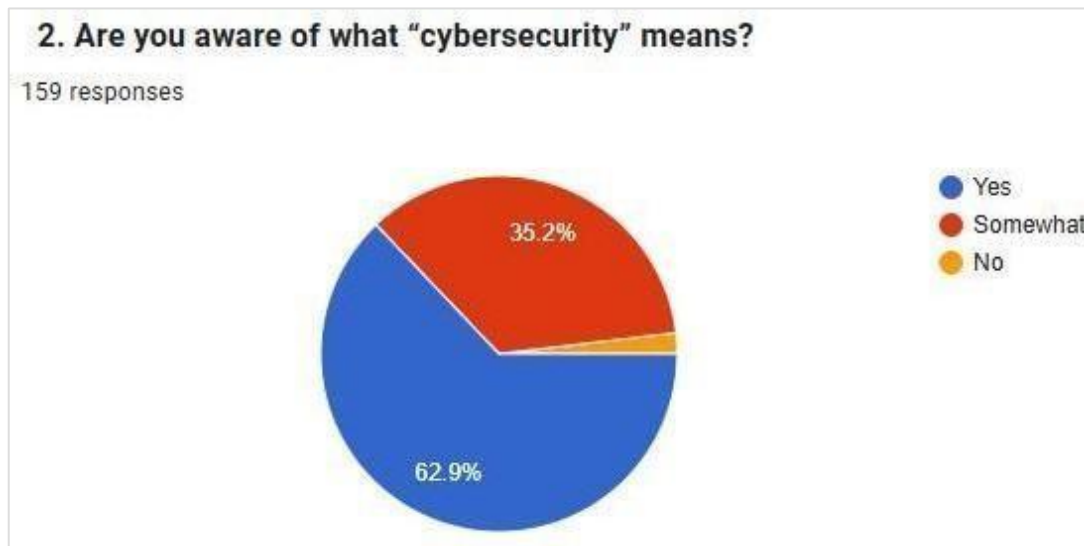
Pie Chart :



5. Awareness :

Awareness Level	Count (N=159)	Percentage
Yes	100	62.9%
Somewhat	56	35.2%
No	3	1.9%

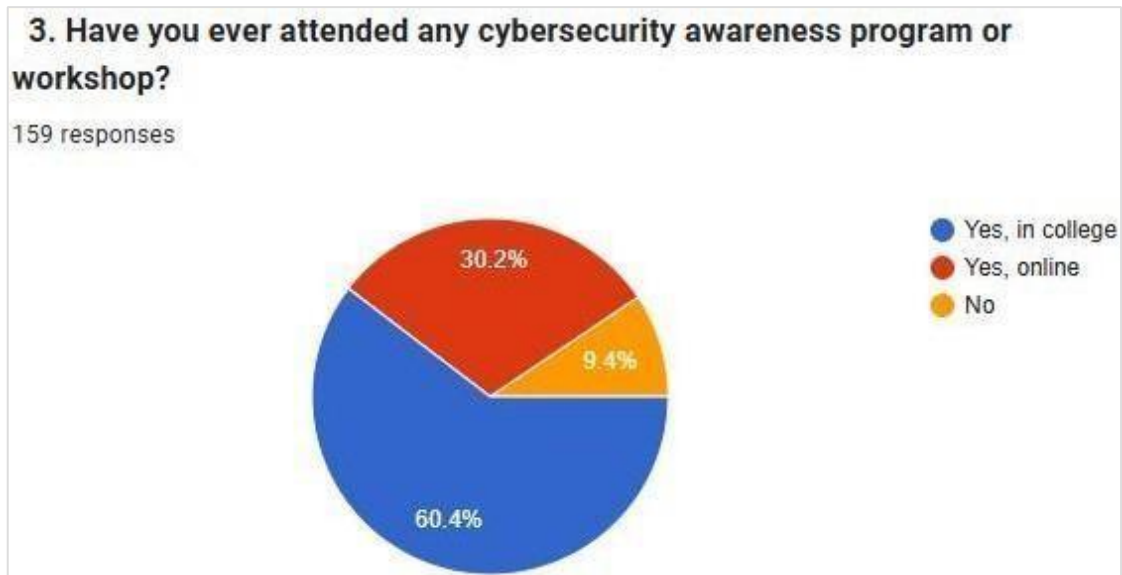
Pie Chart :



6. Training :

Training Status	Count (N=159)	Percentage
Yes, in college	96	60.4%
Yes, online	48	30.2%
No	15	9.4%

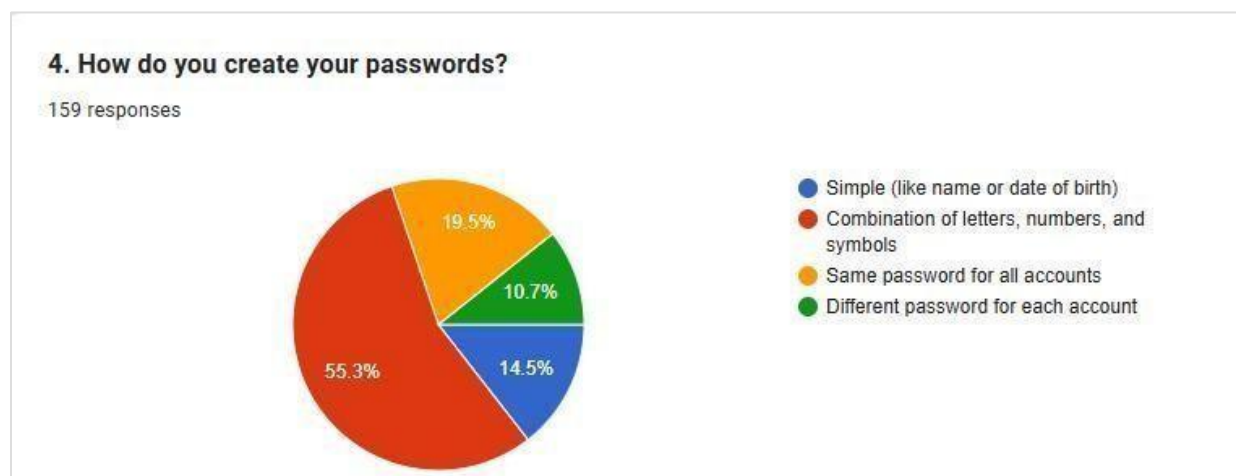
Pie Chart :



7. Password Practice :

Password Practice	Count (N=159)	Percentage
Combination of letters, numbers, and symbols	88	55.3%
Same password for all accounts	31	19.5%
Simple (like name or date of birth)	23	14.5%
Different password for each account	17	10.7%

Pie Chart :



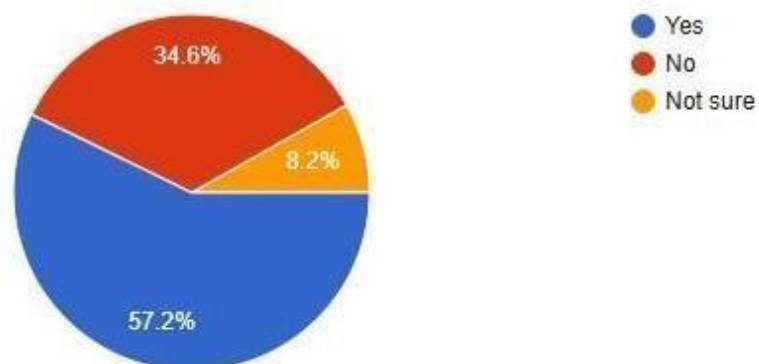
8. Experience :

Experience	Count (N=159)	Percentage
Yes	91	57.2%
No	55	34.6%
Not sure	13	8.2%

Pie Chart :

5. Have you ever experienced or known someone who faced a cyberattack (like hacking or phishing)?

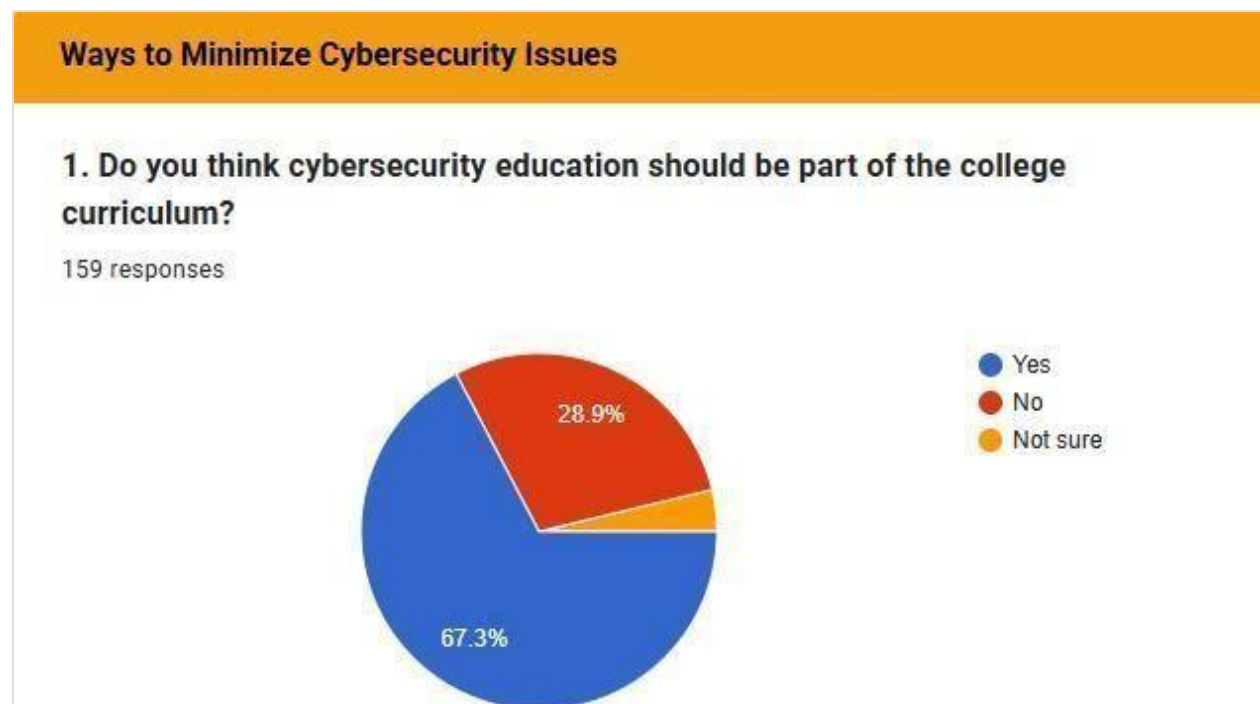
159 responses



9. Opinion :

Opinion	Count (N=159)	Percentage
Yes	107	67.3%
No	46	28.9%
Not sure	6	3.8%

Pie Chart :



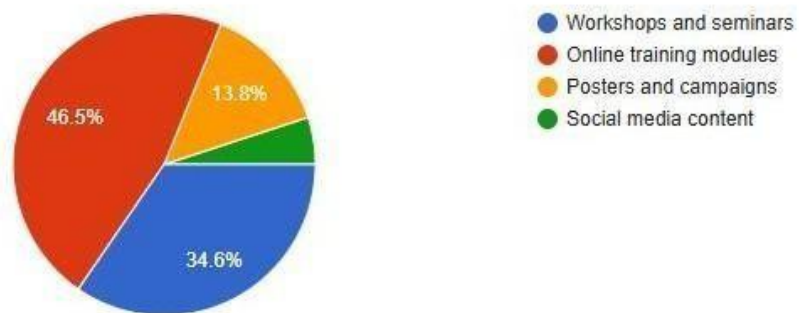
10. Preferred Method :

Preferred Method	Count (N=159)	Percentage
Online training modules	74	46.5%
Workshops and seminars	55	34.6%
Posters and campaigns	22	13.8%
Social media content	8	5.0%

Pie Chart :

2. Which method do you think is best for spreading cybersecurity awareness?

159 responses



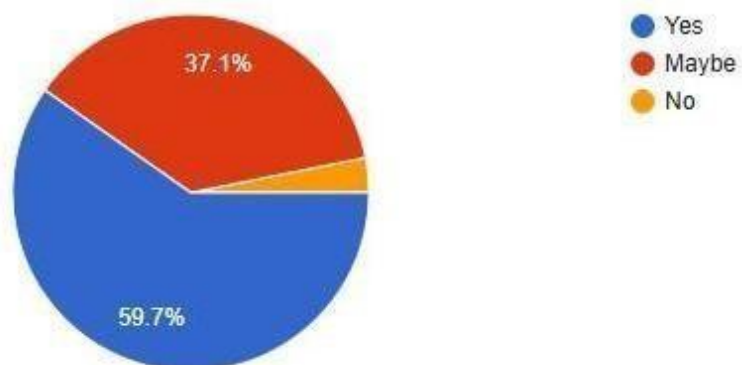
11. Willingness :

Willingness to Participate	Count (N=159)	Percentage
Yes	95	59.7%
Maybe	59	37.1%
No	5	3.1%

Pie Chart :

3. Would you participate in cybersecurity awareness programs if organized in your college?

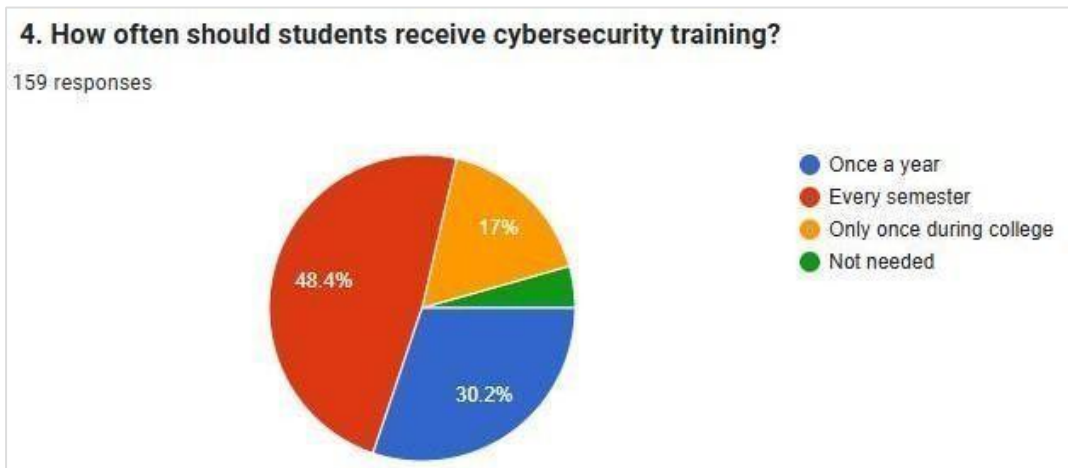
159 responses



12. Frequency :

Preferred Frequency	Count (N=159)	Percentage
Every semester	77	48.4%
Once a year	48	30.2%
Only once during college	27	17.0%
Not needed	7	4.4%

Pie Chart :



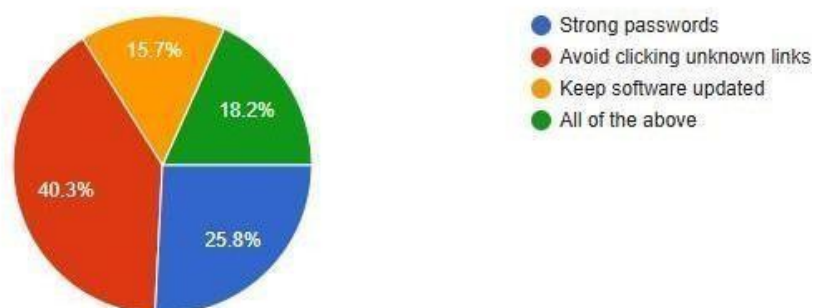
13. Measures :

Perceived Effective Measure	Count (N=159)	Percentage
All of the above	64	40.3%
Avoid clicking unknown links	41	25.8%
Strong passwords	29	18.2%
Keep software updated	25	15.7%

Pie Chart :

5. In your opinion, what is the most effective way to stay safe online?

159 responses



5.2 SAMPLE DEMOGRAPHICS AND USAGE PATTERNS

Demographic Variable	Category	Percentage
Gender	Male	70.4%
Course	MCA	51.6%
Daily Internet Use	Daily	60.4%
	Occasionally	25.8%

5.3 AWARENESS AND TRAINING EXPOSURE

Question	Category	Percentage
Aware of "Cybersecurity"	Yes	62.9%
	Somewhat	35.2%
Attended Awareness Program	Yes, in college	60.4%
	Yes, online	30.2%

- A large majority 62.9% claim subjective awareness of cybersecurity. The combined attendance at college or online training 90.6% suggests local institutions are actively engaging in awareness efforts.

5.4 CYBERSECURITY PRACTICES: THE KNOWLEDGE-BEHAVIOR GAP

The data provides critical evidence of risky behaviors, despite reported awareness.

Question	Category	Percentage	Analysis
How do you create your passwords?	Combination of letters, numbers, and symbols	55.3%	Positive Practice
	Same password for all accounts	19.5%	High Risk Behavior
	Simple (e.g., name/DOB)	14.5%	High Risk Behavior
Experienced a Cyberattack?	Yes	57.2%	High Exposure Validation

- **Core Conflict:** A critical 34.0% (19.5% + 14.5%) of students admit to engaging in unsafe password practices, providing empirical evidence of the Knowledge-Behavior Gap identified in the literature.
- **Exposure:** The finding that 57.2% of students have been directly impacted or know someone who has faced a cyberattack validates the high-risk environment.

5.5 FUTURE TRAINING STRATEGIES AND DISCUSSION

Question	Category	Percentage
Education in College Curriculum?	Yes	67.3%
How Often to Train?	Every semester	48.4%
Best Spreading Method?	Online training modules	46.5%
Would you participate?	Yes	59.7%

- **Policy Demand:** Students overwhelmingly support integrating cybersecurity education into the college curriculum 67.3% and demand frequent training, with 48.4% preferring it every semester.
- **Pedagogy:** Students favor interactive methods (Online Modules 46.5% and Workshops 34.6%, confirming the need to shift to experiential, hands-on learning.

6 FINDINGS AND SUGGESTIONS

6.1 SUMMARY OF KEY FINDINGS AND CONCLUSION

The analysis confirms the fundamental conflict between awareness and practice in the local academic environment.

Hypothesis	Conclusion	Justification
Null Hypothesis (H0): No significant level of awareness or difference.	Rejected	High awareness 62.9% and high cyberattack exposure 57.2% are significant. The failure of H0 is justified by the significant quantification of unsafe practices 34.0% unsafe passwords.
Alternative Hypothesis (H1): Significant level of awareness and/or difference.	Supported	The data supports H1 by confirming high awareness, but the existence and quantification of the Knowledge-Behavior Gap confirms the need for major policy intervention.

6.2 SUGGESTIONS AND ACTIONABLE RECOMMENDATIONS

The following recommendations are provided to address the Knowledge-Behavior Gap and enhance security practices.

Recommendation 1: Mandate Continuous, Semester-Based Behavioral Training

- **Strategy:** Shift from one-time awareness workshops to mandatory, semester-based Cyber Hygiene Modules integrated into the curriculum.
- **Action Required:**
 - **Frequency:** Training must be conducted every semester 48.4% student demand to reinforce habits.
 - **Focus:** Modules must explicitly target the highest failure points: Phishing Recognition and Password Management.

Recommendation 2: Adopt Experiential Learning and Simulation Pedagogy

- **Strategy:** Utilize the student preference for interactive formats 81.1% to close the knowledge-behavior gap.

- **Action Required:**
 - Conduct regular, controlled Phishing Simulation Exercises to test and train real-world recognition skills.
 - Develop Online Modules that use gamification or interactive tasks to promote the use of Password Managers, directly combating the 34.0% unsafe password behavior.

Recommendation 3: Institutionalize Secure Password Practice

- **Strategy:** Enforce and facilitate a "Different Password for Each Account" standard to eliminate high-risk behavior.

- **Action Required:**
 - **Policy:** Mandate a policy for secure, multi-factor authentication for all college-related digital services.
 - **Training:** Integrate mandatory training on the secure use of Password Manager **Tools** into the curriculum to make secure practices easy and habitual.

7 FUTURE SCOPE

Policy Validation:

One of the most significant areas for future work involves the validation of proposed interventions and policies. Future researchers should design and conduct controlled experimental studies to test the real-world effectiveness of the suggested semester-wise cybersecurity training modules. Such longitudinal research can evaluate whether consistent awareness sessions genuinely lead to measurable reductions in unsafe digital practices, such as password sharing, falling for phishing attempts, or neglecting software updates. The outcomes of these experiments will provide empirical evidence that can inform both institutional and governmental cybersecurity education policies.

Comparative Analysis:

Another promising direction is to expand the research to larger and more diverse populations. While the current study focuses on students in the Varale area, future research can include participants from multiple regions, cultural backgrounds, and educational settings. Comparative studies between schools, colleges, and universities across both urban and rural environments would provide a clearer understanding of how demographic and infrastructural factors influence cybersecurity awareness. Additionally, such large-scale comparisons could help identify best practices that can be adapted and implemented in different educational contexts.

Technological Focus:

The integration of technology into cybersecurity education presents yet another valuable opportunity for future exploration. Researchers should examine the potential of mobile applications, gamified learning platforms, and interactive e-learning modules as innovative tools for promoting cybersecurity awareness among students and the wider community. These technological solutions can provide engaging, self-paced, and practical learning experiences that enhance user retention and behavioral change. Further investigation into how such tools can be customized for specific age groups or educational levels would also contribute to more effective digital literacy programs.

8 LIMITATIONS OF STUDY

8.1 LIMITED SCOPE AND GENERALIZABILITY

While the findings of this research provide meaningful insights into the cybersecurity awareness levels of college students in the Varale area, it is important to acknowledge certain limitations that may affect the generalizability of the results. The study was conducted within a specific geographical boundary, focusing primarily on undergraduate and postgraduate students enrolled in colleges located in and around Varale. Consequently, the outcomes may not fully capture the diversity of experiences, technological exposure, and educational practices present among students in other regions or institutions across the country.

The demographic characteristics of the selected participants such as their access to digital infrastructure, institutional support, and socioeconomic background could also have influenced their levels of cybersecurity knowledge and behavior. Students in metropolitan or technology-oriented educational environments, for example, might display significantly higher awareness compared to those in smaller towns or rural settings. Therefore, while the study successfully identifies key trends and issues within the local context, caution must be exercised when attempting to generalize these findings to a national or international scale.

To enhance generalizability, future research should include larger and more diverse sample groups, covering different geographic regions, academic disciplines, and institutional types. A comparative approach involving multiple colleges or universities would enable a broader understanding of cybersecurity awareness patterns and allow researchers to identify region-specific gaps and best practices. Expanding the sample size and scope in future studies will help ensure that the conclusions drawn are more representative of the overall student population and can be applied to policy-making and educational interventions at a wider level.

8.2 RELIANCE ON SELF-REPORTED PERCEPTIONS

Another important limitation of this study lies in its dependence on self-reported data obtained through the structured questionnaire. Since the primary data collection method involved asking participants to evaluate and describe their own cybersecurity practices, perceptions, and levels of awareness, the responses are inherently subjective. Self-reporting relies on the participants' honesty, memory, and self-perception, all of which can introduce varying degrees of inaccuracy or bias into the results.

One significant concern associated with self-reported data is the Social Desirability Bias a psychological tendency among respondents to present themselves in a more favorable light. In this context, participants may have overstated their cybersecurity awareness or underreported risky online behaviors to align with socially accepted norms or to appear more knowledgeable and responsible. As a result, the true extent of unsafe digital practices among students could be slightly higher than what the data suggests.

Furthermore, self-reported responses can also be influenced by factors such as misunderstanding of questions, selective recall, or lack of self-awareness regarding one's digital habits. For instance, a student might claim to use strong passwords or regularly update their devices but may not actually follow these practices consistently in real life. Such discrepancies between reported and actual behavior can affect the overall accuracy of the findings.

To address this limitation, future research should incorporate complementary data collection methods such as observational studies, simulated phishing exercises, or digital behavior tracking (conducted ethically and with proper consent). These approaches can help validate self-reported responses and provide a more objective measure of students' actual cybersecurity practices. Combining self-report data with behavioral assessments would lead to a more comprehensive and accurate understanding of the awareness-to-action gap in cybersecurity among students.

8.3 CONSTRAINTS

Every research study operates within certain practical boundaries, and this study is no exception. The most significant limitations encountered during the research process were related to time and resource constraints, which inevitably influenced the depth and scope of the analysis. Although the study successfully met its primary objectives, these limitations restricted the researcher's ability to conduct more advanced analyses and integrate additional data sources that could have strengthened the findings.

Due to the limited duration of the research period, it was not possible to incorporate objective academic records or performance indicators to establish a stronger link between cybersecurity awareness and students' actual academic or behavioral outcomes. Accessing and analyzing such institutional data would have required additional permissions, extended timelines, and larger-scale coordination with multiple colleges. As a result, the study relied primarily on self-reported information rather than verified academic performance or administrative records. While this approach was practical and time-efficient, it limited the capacity to measure the real-world impact of cybersecurity knowledge on students' digital conduct or academic engagement.

In addition to time limitations, resource constraints including limited access to advanced analytical tools and statistical software prevented the use of more sophisticated quantitative methods, such as inferential or multivariate statistical analyses. The study primarily utilized descriptive statistics to summarize and interpret the collected data. Although these methods were sufficient to identify patterns and trends, they do not allow for the establishment of direct causal relationships between variables. Therefore, the conclusions drawn should be interpreted as indicative rather than definitive evidence of cause and effect.

Future research, with more extensive resources, broader institutional cooperation, and longer timeframes, could overcome these limitations by incorporating mixed-method approaches, including both quantitative and qualitative data, and employing inferential statistical models to test hypotheses more rigorously. Such efforts would enhance the analytical depth and allow for stronger, evidence-based conclusions regarding the factors influencing cybersecurity awareness and behavior among students.

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ANNEXURE

The Annexure contains supplemental materials crucial to the research study, including the full research instrument used for primary data collection.

Research Instrument: Questionnaire

Questionnaire Title: A Study on Cybersecurity Awareness among Undergraduate and Postgraduate College Students (Varale Area)

Section 1: Personal Information

1. Name:
2. Gender: Male / Female
3. Age:
4. Education Level: Undergraduate / Postgraduate
5. College Name:
6. Course: MCA / MBA / Engineering / Other

Section 2: Cybersecurity Awareness (Present State)

1. How often do you use the internet for study or personal work?
(a) Daily (b) Occasionally (c) Rarely (d) Never
2. Are you aware of what “cybersecurity” means?
(a) Yes (b) Somewhat (c) No
3. Have you ever attended any cybersecurity awareness program or workshop?
(a) Yes, in college (b) Yes, online (c) No
4. How do you create your passwords?
(a) Simple (like name or date of birth) (b) Combination of letters, numbers, and symbols (c) Same password for all accounts (d) Different password for each account
5. Have you ever experienced or known someone who faced a cyberattack (like hacking or phishing)?
(a) Yes (b) No (c) Not sure

Section 3: Future — Ways to Minimize Cybersecurity Issues

1. Do you think cybersecurity education should be part of the college curriculum?
(a) Yes (b) No (c) Not sure
2. Which method do you think is best for spreading cybersecurity awareness?
(a) Workshops and seminars (b) Online training modules (c) Posters and campaigns (d) Social media content
3. Would you participate in cybersecurity awareness programs if organized in your college?
(a) Yes (b) Maybe (c) No
4. How often should students receive cybersecurity training?
(a) Once a year (b) Every semester (c) Only once during college (d) Not needed
5. In your opinion, what is the most effective way to stay safe online?
(a) Strong passwords (b) Avoid clicking unknown links (c) Keep software updated (d) All of the above